

## 23.2 The wild bootstrap

The residual bootstrap keeps the design but moves every residual to a random position, destroying any link between the size of an error and where it occurred. The case bootstrap keeps that link but changes the design. The **wild bootstrap** keeps both: each residual stays with its own row of  $\mathbf{X}$ , and only a random multiplier changes (Wu 1986; Liu 1988; Mammen 1993).

### Definition 23.2.1. Wild bootstrap

Let  $\mathbf{X}$  have rank  $p$ , with least squares estimate  $\hat{\boldsymbol{\beta}}$ , residuals  $\hat{\varepsilon}_i$  and leverages  $h_{ii} < 1$ . Choose transformed residuals  $u_i$ , one of

$$u_i = \hat{\varepsilon}_i, \quad u_i = \frac{\hat{\varepsilon}_i}{\sqrt{1 - h_{ii}}}, \quad u_i = \frac{\hat{\varepsilon}_i}{1 - h_{ii}},$$

and a law for the **multipliers**, with mean 0 and variance 1. The wild bootstrap draws  $v_1, \dots, v_n$  independently from that law, independently of the data, sets

$$Y_i^* = \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}} + u_i v_i, \quad i = 1, \dots, n,$$

and computes  $\hat{\boldsymbol{\beta}}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}^*$ . Two standard multiplier laws are the **Rademacher** law,  $v = \pm 1$  with probability  $1/2$  each, and **Mammen's** two-point law

$$\Pr\left(v = \frac{1 - \sqrt{5}}{2}\right) = \frac{\sqrt{5} + 1}{2\sqrt{5}}, \quad \Pr\left(v = \frac{1 + \sqrt{5}}{2}\right) = \frac{\sqrt{5} - 1}{2\sqrt{5}}.$$

In the bootstrap world the error of case  $i$  is  $u_i v_i$ , with mean zero and its own variance  $u_i^2$ . One residual is a poor estimate of a variance, but  $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$  needs only a weighted average of the variances, as for the sandwich.

### Proposition 23.2.1. Moments of the wild bootstrap

In the wild bootstrap, with  $\mathbf{U} = \text{diag}(u_1, \dots, u_n)$  and  $\mathbf{a} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}$ :

(a)  $E_*(\hat{\boldsymbol{\beta}}^*) = \hat{\boldsymbol{\beta}}$  and

$$\text{Cov}_*(\hat{\boldsymbol{\beta}}^*) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{U}^2 \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}, \quad (23.2.1)$$

which is the sandwich estimator HC0, HC2 or HC3 (Theorem 21.4.2) for the three choices of  $u_i$ ;

(b)  $E_*[(\mathbf{c}^\top \hat{\boldsymbol{\beta}}^* - \mathbf{c}^\top \hat{\boldsymbol{\beta}})^3] = E(v^3) \sum_i a_i^3 u_i^3$ .

*Proof.*  $\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{U} \mathbf{v}$ , where  $\mathbf{v} = (v_1, \dots, v_n)^\top$  has mean  $\mathbf{0}$  and covariance  $\mathbf{I}$ . So  $E_*(\hat{\boldsymbol{\beta}}^*) = \hat{\boldsymbol{\beta}}$  and  $\text{Cov}_*(\hat{\boldsymbol{\beta}}^*) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{U} \mathbf{U} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$ . For (b),  $\mathbf{c}^\top (\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}}) = \sum_i a_i u_i v_i$  is a sum of independent terms with mean zero, and the third moment of such a sum is the sum of the third moments, because every cross term contains some  $v_i$  to the first power. ■

So the wild bootstrap reproduces the sandwich exactly, not only to first order as the case

bootstrap does (Proposition 23.1.5). Part (b) explains the choice of multipliers. The true third moment is  $\sum_i a_i^3 E(\varepsilon_i^3)$ , not zero for skewed errors. Rademacher multipliers have  $E(v^3) = 0$  and give a symmetric bootstrap distribution; Mammen's law is the two-point law with third moment 1 (Exercise (B1)) and carries the skewness of the residuals. Davidson and Flachaire (2008) nonetheless favour Rademacher weights for tests, because matching the third moment costs accuracy in the fourth.

## 23.2.1 Consistency under heteroscedasticity

Take the fixed designs of Section 23.1, with contrasts  $\mathbf{c}_n$ , weights  $\mathbf{a}_n$ ,  $w_i = a_{ni}/\|\mathbf{a}_n\|$  and Eq. 23.1.1. The errors  $\varepsilon_{n1}, \dots, \varepsilon_{nn}$  are independent with mean zero and variances  $\sigma_{ni}^2$ , and for some constants  $0 < \underline{\sigma}^2$  and  $K < \infty$ ,

$$\sigma_{ni}^2 \geq \underline{\sigma}^2, \quad E\varepsilon_{ni}^4 \leq K \quad \text{for all } n, i. \quad (23.2.2)$$

The variance of the standardized contrast is  $\tau_n^2 = \sum_i w_i^2 \sigma_{ni}^2$ , which lies between  $\underline{\sigma}^2$  and  $\sqrt{K}$ , since  $\sigma_{ni}^2 \leq (E\varepsilon_{ni}^4)^{1/2}$ .

### Theorem 23.2.2. Consistency of the wild bootstrap

Under Eq. 23.1.1 and Eq. 23.2.2, let  $\hat{\boldsymbol{\beta}}^*$  come from the wild bootstrap with  $u_i = \hat{\varepsilon}_i$  and multipliers from any fixed law with mean 0 and variance 1, and let  $G_n$  and  $\hat{G}_n$  be defined as in Theorem 23.1.4, with this  $\hat{\boldsymbol{\beta}}^*$ . Then  $\sup_x |\hat{G}_n(x) - G_n(x)| \rightarrow 0$  in probability, and both are uniformly close to  $\Phi(x/\tau_n)$ .

*Proof.* Write  $\mathbf{d} = \mathbf{M}\boldsymbol{\varepsilon}$ , so that  $\hat{\varepsilon}_i = \varepsilon_i - d_i$ , and put  $m_n = \max_i |w_i|$ . Four steps.

*Step 1: the bootstrap variance.* Let  $S_n^2 = \sum_i w_i^2 \varepsilon_i^2$ , the bootstrap variance of  $\mathbf{c}_n^\top (\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}})/\|\mathbf{a}_n\|$  by Proposition 23.2.1. We show  $S_n^2 - \tau_n^2 \rightarrow 0$  in probability. Expand  $S_n^2 = \sum_i w_i^2 \varepsilon_i^2 - 2 \sum_i w_i^2 \varepsilon_i d_i + \sum_i w_i^2 d_i^2$ . The first sum has mean  $\tau_n^2$  and variance  $\sum_i w_i^4 \text{Var}(\varepsilon_i^2) \leq Km_n^2 \sum_i w_i^2 = Km_n^2 \rightarrow 0$ . The third is at most  $m_n^2 \|\mathbf{d}\|^2$ , and  $E\|\mathbf{d}\|^2 = \text{tr}(\mathbf{M} \text{Cov}(\boldsymbol{\varepsilon})) \leq p\sqrt{K}$ , so it tends to zero in probability. The middle sum is at most  $2(\sum_i w_i^2 \varepsilon_i^2)^{1/2} (\sum_i w_i^2 d_i^2)^{1/2}$  by Cauchy–Schwarz, a bounded factor times one that tends to zero.

*Step 2: no dominant term.*  $\max_i |w_i \hat{\varepsilon}_i| \leq \max_i |w_i \varepsilon_i| + m_n \|\mathbf{d}\|$ . The second term tends to zero in probability, and for  $\eta > 0$

$$\Pr(\max_i |w_i \varepsilon_i| > \eta) \leq \sum_i \frac{w_i^4 E\varepsilon_i^4}{\eta^4} \leq \frac{Km_n^2}{\eta^4},$$

which tends to zero. So  $\max_i |w_i \hat{\varepsilon}_i| \rightarrow 0$  in probability.

*Step 3: the bootstrap limit.* Given the data,  $U_{ni}^* = w_i \hat{\varepsilon}_i v_i / S_n$  are independent with mean zero and variances summing to one. With  $g(t) = E[v^2 \mathbf{1}\{|v| > t\}]$ , which decreases to zero as  $t \rightarrow \infty$ ,

$$\begin{aligned} L_n^*(\eta) &= \sum_i \frac{w_i^2 \hat{\varepsilon}_i^2}{S_n^2} g\left(\frac{\eta S_n}{|w_i \hat{\varepsilon}_i|}\right) \\ &\leq g\left(\frac{\eta S_n}{\max_i |w_i \hat{\varepsilon}_i|}\right). \end{aligned}$$

By Steps 1 and 2 the argument of  $g$  tends to infinity in probability, because  $S_n^2 \geq \underline{\sigma}^2/2$  with probability tending to one. So  $L_n^*(\eta) \rightarrow 0$  in probability, and Lemma 23.1.2 gives  $\sup_x |\Pr_*(\sum_i w_i \hat{\varepsilon}_i v_i \leq x) - \Phi(x/S_n)| \rightarrow 0$  in probability. Since  $S_n/\tau_n \rightarrow 1$  in probability, the same holds with  $\Phi(x/\tau_n)$ , as at the end of the proof of Theorem 23.1.4.

*Step 4: the true distribution.*  $\mathbf{c}_n^\top(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\|\mathbf{a}_n\| = \sum_i w_i \varepsilon_i$ . With  $U_{ni} = w_i \varepsilon_i/\tau_n$ , Lyapunov's condition holds:  $\sum_i E U_{ni}^4 \leq K m_n^2/\sigma^4 \rightarrow 0$ . So  $\sum_i w_i \varepsilon_i/\tau_n \rightarrow N(0, 1)$ , and by Pólya's theorem  $\sup_x |\Pr(\sum_i w_i \varepsilon_i \leq x) - \Phi(x/\tau_n)| \rightarrow 0$ . The triangle inequality completes the proof. ■

An argument like Step 1 shows why the residual bootstrap fails here. Its variance for the standardized contrast is  $\hat{\sigma}_*^2$ , whose expectation  $n^{-1} \sum_i (1 - h_{ii}) \sigma_{ni}^2$  is an unweighted average of the error variances (Exercise (B1)), not the weighted average  $\tau_n^2$ . So it settles on a normal law with the variance the classical formula gives. The wild bootstrap succeeds because each resampled error keeps the variance of its own case.

### Example. The wild bootstrap for Engel's data

For the line of Example (Engel's food expenditure), 4000 wild resamples with  $u_i = \hat{\varepsilon}_i$  give standard errors 0.05119 (Rademacher) and 0.05217 (Mammen), Monte Carlo estimates of the HC0 value 0.05177. With  $u_i = \hat{\varepsilon}_i/(1 - h_{ii})$  they give 0.06653, estimating HC3, 0.06639.

The shape is more interesting. The wild distribution in Figure 23.1.1(b) has two modes, at 0.439 and 0.531. The cause is the richest household, whose leverage 0.255 was examined in Chapter 20 and whose share of the HC3 variance was measured in Chapter 21. Its term  $a_i^2 \hat{\varepsilon}_i^2$  accounts for a fraction 0.78 of the HC0 variance  $\sum_j a_j^2 \hat{\varepsilon}_j^2$ , so the condition behind Step 2 of the proof is badly violated. The bootstrap slope is dominated by the contribution  $a_i \hat{\varepsilon}_i v_i$  of this one household, which with Rademacher multipliers is  $\pm 0.046$ , and the modes sit at  $\hat{\beta}_1 \mp 0.046$ . The sampling distribution of the slope is dominated by that household's error too, and one observation cannot reveal the law it came from; the remedy is the one given in those chapters.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.engel.load_pandas().data
y = data["foodexp"].to_numpy()
x = data["income"].to_numpy()
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape
XtX_inv = np.linalg.inv(X.T @ X)
beta_hat = XtX_inv @ X.T @ y
e_hat = y - X @ beta_hat
h = np.sum((X @ XtX_inv) * X, axis=1) # leverages
rng = np.random.default_rng(2301)
B = 4000
A = XtX_inv @ X.T # beta_hat = A y
fit = X @ beta_hat

golden = (1 + np.sqrt(5)) / 2
mammen_values = np.array([1 - golden, golden]) # -0.618 and 1.618
mammen_probs = np.array([golden / np.sqrt(5), 1 - golden / np.sqrt(5)])

def wild_bootstrap(resid, B, weights="rademacher"):
```

```

    """Slopes from  $y_i = x_i' \beta + \varepsilon_i$ , with independent multipliers
     $v_i$ ."""
    if weights == "rademacher":
        v = rng.choice([-1.0, 1.0], size=(B, n))
    else:
        v = rng.choice(mammen_values, p=mammen_probs, size=(B, n))
    y_star = fit + resid * v
    return (y_star @ A.T)[: , 1]

slopes_wild = wild_bootstrap(e_hat, B)
slopes_mammen = wild_bootstrap(e_hat, B, weights="mammen")
slopes_wild3 = wild_bootstrap(e_hat / (1 - h), B)
se_hc0 = np.sqrt(A[1] ** 2 @ e_hat ** 2)
se_hc3 = np.sqrt(A[1] ** 2 @ (e_hat / (1 - h)) ** 2)
print(f"wild se: Rademacher {slopes_wild.std():.5f}, Mammen {slopes_mammen.std():.5f}
      ", "
      f"HC3-type {slopes_wild3.std():.5f}")
print(f"sandwich se: HC0 {se_hc0:.5f}, HC3 {se_hc3:.5f}")

```

[Run this code online](#)

## 23.2.2 Bootstrap or sandwich?

For a standard error the wild bootstrap adds nothing to the sandwich it reproduces. It earns its place through the *studentized* statistic  $(\mathbf{c}^\top \hat{\boldsymbol{\beta}} - \mathbf{c}^\top \boldsymbol{\beta}) / \widehat{\text{se}}_{\text{HC}}$ , whose noisy denominator makes its small-sample distribution far from  $t$  when some leverages are large; the simulations of [Section 23.3](#) show the wild bootstrap- $t$  beating the HC2  $t$  interval. For a test of  $\mathbf{c}^\top \boldsymbol{\beta} = d$ , the resamples should come from the restricted fit, so that the bootstrap approximates the null distribution even when the null hypothesis is false (Davidson and Flachaire 2008).

## 23.2.3 Exercises

### 23.2.3.1 A. Check your understanding

#### Exercise. A1

With Rademacher multipliers and  $u_i = \hat{\varepsilon}_i$ , how many distinct wild resamples are there? Show that the bootstrap variance of  $\mathbf{c}^\top \hat{\boldsymbol{\beta}}^*$  computed by averaging over all of them equals the HC0 variance exactly.

*Solution.* There are  $2^n$  sign vectors, equally likely. Averaging  $(\sum_i a_i \hat{\varepsilon}_i v_i)^2$  over them, every cross term  $a_i a_j \hat{\varepsilon}_i \hat{\varepsilon}_j v_i v_j$  with  $i \neq j$  averages to zero, because half of the sign vectors give  $v_i v_j = 1$  and half give  $-1$ . What remains is  $\sum_i a_i^2 \hat{\varepsilon}_i^2$ , the  $(\mathbf{c}, \mathbf{c})$  entry of HC0. The bootstrap mean is  $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$ , so this is the variance.

#### Exercise. A2

Two groups of sizes  $n_1$  and  $n_2$  are compared through the difference of their means, fitted by least squares with one indicator column per group. Show that the wild bootstrap with  $u_i = \hat{\varepsilon}_i$  gives the variance  $\hat{\sigma}_1^2/n_1 + \hat{\sigma}_2^2/n_2$ , where  $\hat{\sigma}_k^2$  is the within-group variance with divisor

$n_k$ , and that  $u_i = \hat{\varepsilon}_i / \sqrt{1 - h_{ii}}$  gives the Welch variance  $s_1^2/n_1 + s_2^2/n_2$ .

### 23.2.3.2 B. Practice

#### Exercise. B1

Let  $v$  take two values  $a < 0 < b$ . Show that  $E(v) = 0$  and  $E(v^2) = 1$  force  $ab = -1$  and  $\Pr(v = a) = b/(b - a)$ , and that then  $E(v^3) = a + b$ . Deduce that Mammen's law of [Definition 23.2.1](#) is the only two-point law with mean 0, variance 1 and third moment 1, and compute its fourth moment.

*Solution.* With  $q = \Pr(v = a)$ ,  $qa + (1 - q)b = 0$  gives  $q = b/(b - a)$ . Then  $E(v^2) = (ba^2 - ab^2)/(b - a) = -ab$ , so  $ab = -1$ . Similarly  $E(v^3) = (ba^3 - ab^3)/(b - a) = -ab(a + b) = a + b$ . Third moment one means  $a + b = 1$  and  $ab = -1$ , so  $a$  and  $b$  are the roots of  $t^2 - t - 1 = 0$ , namely  $(1 \mp \sqrt{5})/2$ , and  $q = b/(b - a) = (1 + \sqrt{5})/(2\sqrt{5})$ . The fourth moment is  $(ba^4 - ab^4)/(b - a) = -ab(a^2 + ab + b^2) = (a + b)^2 - ab = 2$ , compared with 1 for Rademacher weights and 3 for the normal.

### 23.2.3.3 C. Going deeper

#### Exercise. C1

Suppose the cases fall into clusters and errors are correlated within clusters but independent between them. Show that the wild bootstrap of [Definition 23.2.1](#) gives the HC0 covariance, which ignores the correlation. Propose a version in which all cases in a cluster share one multiplier, and show that its bootstrap covariance is the cluster-robust sandwich  $(\mathbf{X}^\top \mathbf{X})^{-1} \sum_g \mathbf{X}_g^\top \hat{\varepsilon}_g \hat{\varepsilon}_g^\top \mathbf{X}_g (\mathbf{X}^\top \mathbf{X})^{-1}$ .